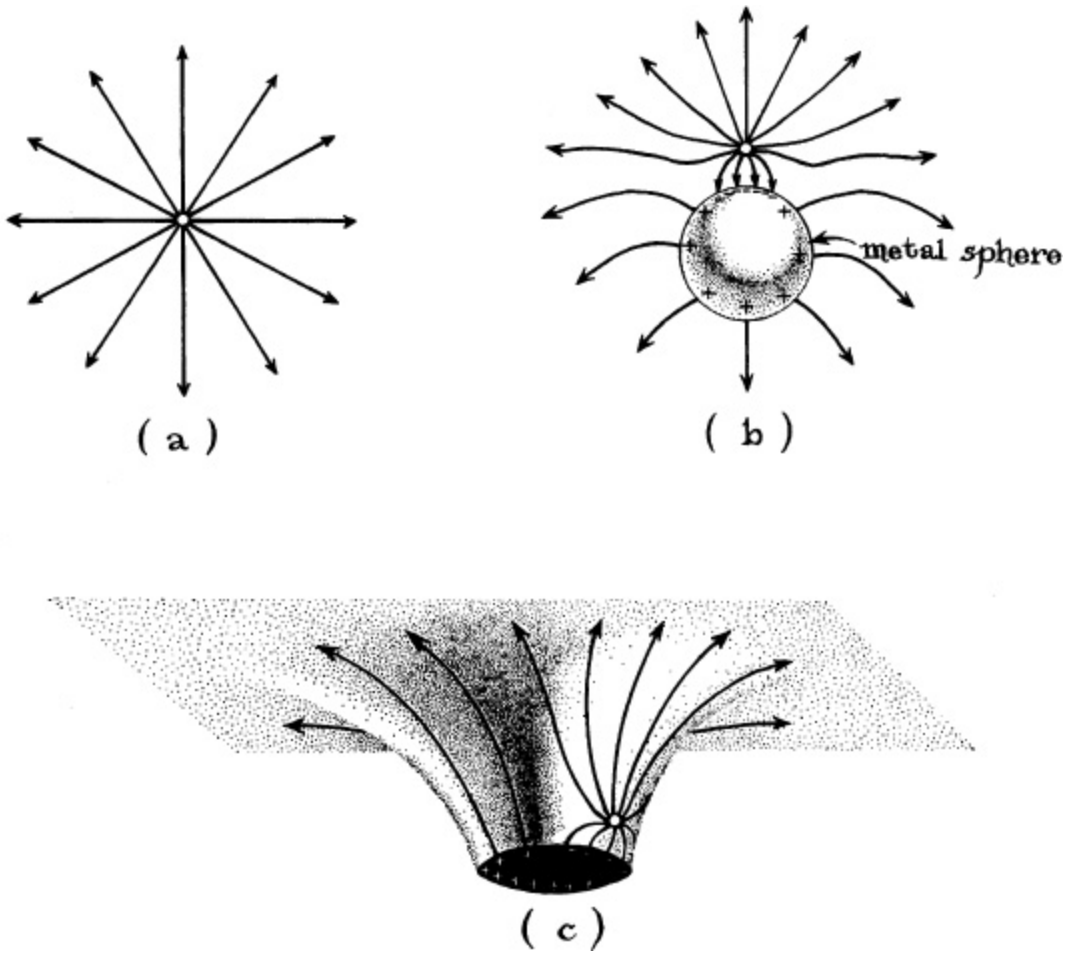


11.3 (a) The electric field lines produced by a positively charged metal pellet at rest, alone, in flat spacetime. (b) The electric field lines when the pellet is at rest just above an electrically conducting, metal sphere in flat spacetime. The pellet's electric field polarizes the sphere. (c) The electric field lines when the pellet is at rest just above the horizon of a black hole. The pellet's electric field appears to polarize the horizon.

In 1971 Hanni and Ruffirti, and independently Robert Wald of Princeton University and Jeff Cohen of the Princeton Institute for Advanced Study, computed the shapes of the electric field lines produced by a charged pellet near the horizon of a nonspinning black hole. Their computations, based on the standard curved spacetime paradigm, revealed that the curvature of spacetime distorts the field lines in the manner shown in Figure 11.3c. Hanni and Ruffini, noticing the similarity to the field lines in Figure 11.3b [look at diagram (c) from below, and it will be nearly the same as diagram (b)], suggested that we can think of a black hole's horizon in the same manner as we think of a metal sphere; that is, we can regard the horizon as a thin membrane composed of positively and negatively charged particles, a membrane similar to the sphere's metal. Normally there are equal numbers